

Mitochondria Metabolic Modeling : Neortex development

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Abstract

During neural development, the neocortex grows through the coordinated action of progenitor cells and neurons, two cell types with very different energy needs. While we know that oxygen levels are different across the developing brain, we don't fully understand how this affects the metabolic pathways of different cell types. I investigated mitochondrial metabolism in four cell types of the developing neocortex (E16.5) apical progenitors, basal progenitors, deep layer neurons, and upper layer neurons, using flux balance analysis (FBA) integrated with gene expression data and physiologically almost realistic oxygen constraints.

The results showed a clear metabolic difference between progenitors and neurons. Progenitors make most of their ATP from the TCA cycle (47-57%) and glycolysis (28-29%), with very little from OXPHOS (15-24%). They also synthesize ATP in the mitochondria and transport it to the cytosol using the ATP/ADP carrier. Neurons do the opposite: they shift toward OXPHOS, getting 40-45% of their ATP from the electron transport chain while reducing TCA cycle activity.

The key finding is that oxygen availability is the main factor controlling this metabolic shift. Progenitors have a functional electron transport chain but cannot use it efficiently in their hypoxic environment, so they rely on oxygen-independent pathways instead. This metabolic flexibility allows progenitors to survive and proliferate in low oxygen zones while neurons in more oxygenated layers can use the most efficient ATP production pathway. This work shows that constraint-based metabolic modeling reveals how the oxygen gradient in the developing brain shapes cell-type-specific energy strategies during neural development.

1 Introduction

1.1 The biological context:

1.1.1 Mitochondria and cellular energy metabolism

Mitochondria are organelles found in all eukaryotic cells, they have many essential functions in the cell, the most important of which is ATP production, which is the cell's main source of energy [Chandel \(2014\)](#). Structurally, they have two main compartments: the matrix and the intermembrane space. They also possess a double membrane. The outer membrane which is fairly permeable, and the inner membrane which is folded inward forming structures called cristae [Anderson et al. \(1981\)](#). These folds increase the available surface area, which matters because most of the energy-related reactions happen at this membrane. The region inside the inner membrane is the matrix, where the TCA cycle takes place, and the region between the two membranes is the intermembrane space, which is where protons accumulate to drive ATP production.

One of the main fueling molecules for feeding the TCA cycle and the electron transport chain for ATP production is Acetyl-CoA. There are several pathways that can produce this compound. The main one is glycolysis which happens in the cytosol. Glucose molecules get broken into Pyruvate. Pyruvate is then transported across the inner mitochondrial membrane into the matrix via the pyruvate carrier. Once in the matrix, pyruvate is converted to Acetyl-CoA by an enzyme called pyruvate dehydrogenase. The second main pathway is through beta oxidation of fatty acids. ATP synthesis, with a positive balance of ATP, can be produced by two reactions within mitochondria:

The tricarboxylic acid cycle (TCA): the TCA cycle takes place in the mitochondrial matrix. It is called a cycle because it starts and ends with the same molecule Oxaloacetate. Acetyl-CoA joins Oxaloacetate to form citrate, which then undergoes a series of enzymatic reactions. Through these reactions, the carbon in acetyl-coA gets completely oxidised to CO_2 . While electrons are captured in the electron carriers NADH and FADH_2 , only one molecule of ATP is produced in this phase (per cycle). These electron carriers then feed into the electron transport chain (ETC) to generate the majority of cellular ATP.

The electron transport chain (ETC): the ETC is a series of 5 main complexes (CI, CII, CIII, CIV, CV), embedded into the mitochondrial inner membrane (the cristae). Their main role is to carry electrons from NADH and FADH_2 down a chain, until they reach the Oxygen as an acceptor. NADH passes electrons to Complex I (CI), while FADH_2 enters Complex II (CII); the ETC complexes then pass electrons through the chain with the help of two co-enzymes CoQ and Cytochrome C. As electrons move through each complex, energy is released and used to pump protons from the matrix into the intermembrane space. CII does not pump protons, which is why FADH_2 -derived electrons generate less ATP than NADH. At the final step, electrons reach O_2 , which accepts them and is reduced to H_2O . By the end, a large proton gradient has accumulated from all the reactions, and this gradient is used by CV to phosphorylate ADP

to ATP and create cellular energy [Mookerjee et al. \(2017\)](#).(Figure 1)

The coupling of the ETC with this chemiosmotic process where the proton gradient drives ATP synthesis is what we call oxidative phosphorylation (OXPHOS).

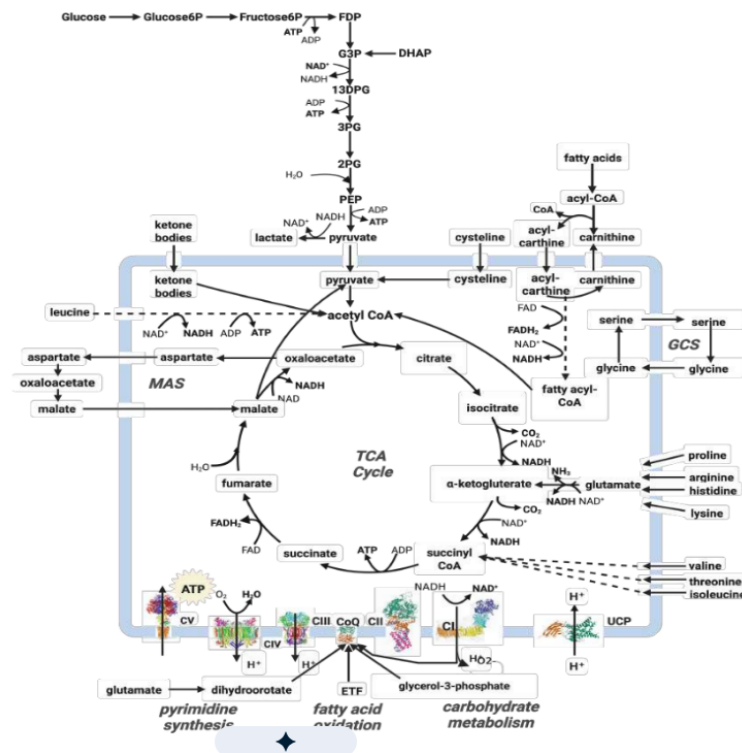


Figure 1: Overview of mitochondrial metabolic pathways. Diagram illustrating the major pathways of mitochondrial metabolism including glycolysis, the TCA cycle, the electron transport chain (CI-CV), fatty acid oxidation, the malate-aspartate shuttle, and associated amino acid and biosynthetic routes modeled in the mitoMammal framework.

1.1.2 Neural development and the developing neocortex

Neurogenesis: the four cell populations

Neurogenesis is the development process where neuronal stem cells divide and differentiate to create neurons [Kriegstein and Alvarez-Buylla \(2009\)](#). In mice, this process starts from E10.5 (embryonic day 10.5) and it takes approximately 8 days [Sun and Hevner \(2014\)](#). Before the whole process starts, the neural tube is still just composed of neuroepithelial cells, which are simple stem cells stacked in a single layer. Those cells need to extend first. Between day 8 and 10, these cells proliferate and divide symmetrically, which we call the expansion phase. When enough cells have been produced, they start dividing asymmetrically so in each division, we will have one daughter that remains a stem cell, and the second one that differentiates and becomes a progenitor. These progenitors then divide further into neurons. With the asymmetric cell divisions, neurogenesis starts.

The early expansion phase creates a “progenitor pool”, in which the number of stem cells determines the number of neurons that could be made. I will first discuss the progenitor pool of cells.

Apical progenitors (AP) reside in the ventricular zone (VZ), the innermost germinal layer in direct contact with the cerebrospinal fluid-filled ventricles. APs are characterized by an elongated morphology with apical and basal processes that extend from the ventricular to the pial surface, spanning the entire thickness of the developing cortex. During the cell cycle, APs undergo interkinetic nuclear migration, wherein the nucleus moves basally during interphase and returns to the apical surface for mitosis, ensuring that cell division occurs exclusively at the ventricular surface. APs express characteristic marker genes including Pax6 and Sox2 transcription factors that maintain their undifferentiated, stem-like state along with GLAST, GFAP, and glutamine synthetase, which reflect their glial-like properties [Götz et al. \(1998\)](#); [Götz and Huttner \(2005\)](#). Following an initial symmetric expansion phase (E10–E12) in which APs double the progenitor pool, they progressively shift toward asymmetric division. During early asymmetric divisions (E12–E13), APs directly generate postmitotic deep-layer neurons (DLNs). As development proceeds (E13–E16), APs increasingly produce basal progenitors (BPs), a secondary progenitor population that detaches from the apical surface to establish the subventricular zone (SVZ). They also divide directly into deep layer neurons.

Basal progenitors (BPs) represent a heterogeneous secondary progenitor population that arises from asymmetric divisions of apical progenitors and migrates outward to form the subventricular zone. BPs are a diverse group: basal radial glia retain attachment to the basal surface and maintain a gene expression profile closely related to their AP origin, accounting for approximately 5% of BPs in mice, while basal intermediate progenitors fully detach from both apical and basal surfaces and constitute the major BP subclass. In contrast to APs, basal intermediate progenitors have severely limited proliferative capacity, typically undergoing only a single cell division before producing two upper-layer neurons (ULN). BPs are characterized by distinct marker gene expression, particularly Tbr2/Eomes (T-box brain protein 2/Eomesodermin), which distinguishes them from APs and postmitotic neurons [Englund et al. \(2005\)](#). By dividing only once and producing two neurons each, BPs amplify neuron production without requiring unlimited AP divisions.

Migration of upper-layer and deep-layer neurons

Deep layer neurons (DLN) come from apical progenitors in the VZ, while upper-layer neurons (ULN) come from basal progenitors in the SVZ. Both types need to migrate toward the pial surface to reach their final positions in the cortex. There are two ways neurons migrate: early on, neurons use somal translocation, they extend a process that attaches to the pial surface while the nucleus moves along this process toward the inside. As the nucleus moves, the process gets shorter but stays attached. Later, neurons switch to gliophilic migration, where they move along the radial fibers of apical progenitors [Nadarajah and Parnavelas \(2002\)](#). DLN, since they're born first, settle near the pial surface and form the first cortical layers. ULNs are born later and migrate past the DLN to reach more superficial positions.

Because early born neurons end up deeper and later born neurons end up closer to the surface, the cortex develops in an inside out pattern [Buchsbaum and Cappello \(2019\)](#).(Figure 2)

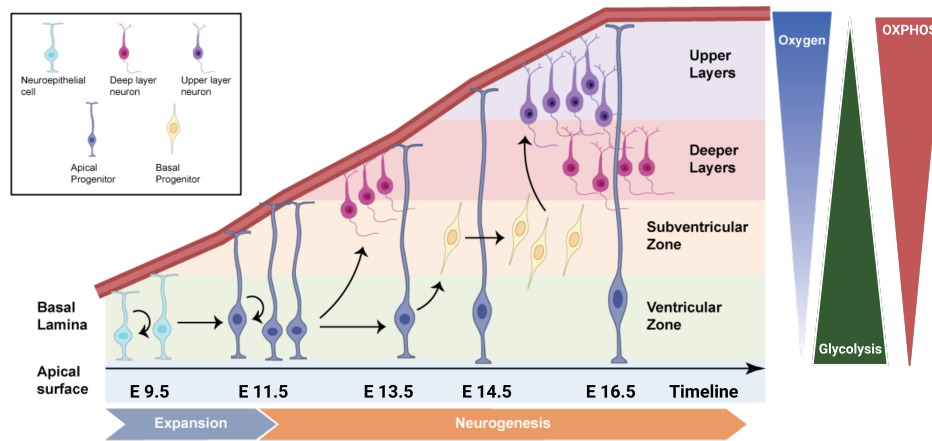


Figure 2: Cortical development timeline and oxygen gradient. Schematic of the developing neocortex showing the temporal progression of cell type generation from E9.5 to E16.5, the spatial organization of the ventricular zone (VZ), subventricular zone (SVZ), and cortical layers, and the opposing gradients of oxygen availability and metabolic strategy (OXPHOS vs. glycolysis) across cortical depth.

1.1.3 Oxygen dynamics in the developing cortex

Blood vessels grow into the developing cortex starting from embryonic day E8.5–E10 [Hogan et al. \(2004\)](#); [Lee et al. \(2001\)](#), growing from the pial surface inward toward the ventricular zone. This creates an oxygen gradient: superficial regions where neurons are located have higher oxygen, while deeper regions where stem cells live stay hypoxic [Dong et al. \(2022\)](#); [Tata et al. \(2015\)](#). Physiological hypoxia is actually necessary for stem cells to stay undifferentiated [Ezashi et al. \(2005\)](#); when oxygen levels increase due to vessel ingrowth, neural progenitors differentiate into neurons [Santilli et al. \(2010\)](#). Evidence from Gpr124 knockout embryos shows that without proper vascularization and oxygen increase, stem cells keep expanding instead of differentiating [Lange et al. \(2016\)](#).

Vascularization begins around E8.5–E10 when a network of blood vessels forms around the neural tube called the perineural vascular plexus. From there, new vessels grow inward from the pial surface toward the ventricular zone, connecting to form an internal network called the subventricular vascular plexus. When blood vessels grow in neurons switch to OXPHOS because oxygen levels are higher now. This allows neurons to produce more ATP for functioning.

1.1.4 Metabolic adaptation during neurogenesis

The oxygen gradient directly controls metabolic pathways. In hypoxia regions where progenitors reside (VZ and SVZ), apical and basal progenitors are expected to rely highly on glycolysis because oxygen is low [Agostini et al. \(2016\)](#); [Zheng et al. \(2016\)](#). When blood vessels grow in, neurons switch to OXPHOS to meet their elevated energy demands [Magistretti and Allaman \(2015\)](#); [Yellen \(2018\)](#).

1.2 The Computational context:

1.2.1 Genome scale metabolic models

Genome scale metabolic models (GEMs) computationally describe a representation of a metabolic network of the cell where gene-protein-reaction (GPR) associations link the reactions to the genome for a set of metabolic genes in an organism [Bordbar et al. \(2014\)](#). A GEM and the GPR rules in the model are created using Genome annotation and experimental data. Using constraint-based modelling techniques such as flux balance analysis (FBA), we can then predict flux values through this network, which is the rate of all reactions or a set of targeted reactions in the model, by optimizing the fluxes towards a specific biological objective (objective function) using linear programming [Kauffman et al. \(2003\)](#). The objective function is a metabolic goal or target that we assume the cell is trying to maximize or minimize, for example ATP production or biomass growth. Omics data can also be integrated in the model to constrain the reaction bounds (limitations) based on gene expression levels, so that the results can be scientifically more accurate and closer to reality. Additionally, models can have multiple cellular compartments so we can specify in which compartment the metabolites are located (cytosol, mitochondria, extracellular space, etc.). This makes the predictions more biologically meaningful.

1.2.2 The stoichiometric matrix

To compute the flux using linear programming, we first need to convert the model into a stoichiometric matrix. This is a table that describes all the reactions in the model, with rows as metabolites (like glucose or atp) and columns as reactions. The numbers in the table tells you how much of each metabolite is consumed or produced in each reaction. If the number is negative it means it is being consumed and if it is positive it means it is being produced. Thus, this is a way to organise all the network information in a format that the computer and the solver can work with.(Figure 3)

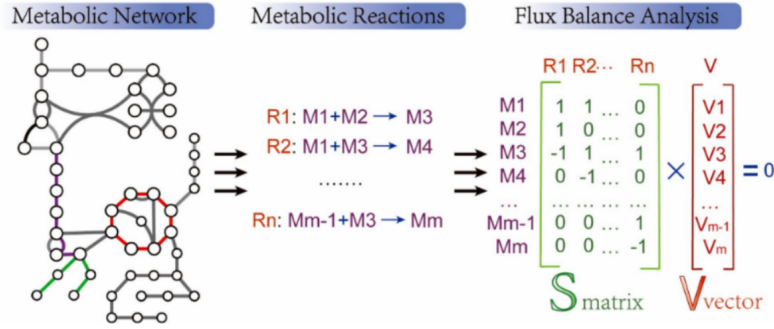


Figure 3: Construction of the stoichiometric matrix for FBA. Diagram showing the conversion of a metabolic network (left) and its set of reactions (center) into the stoichiometric matrix S (right), where each entry represents the stoichiometric coefficient of a metabolite in a given reaction. The matrix equation $S \cdot V = 0$ encodes the steady-state mass balance constraint used in flux balance analysis.

1.2.3 Flux Balance Analysis (FBA)

FBA is an approach that chooses one single metabolic solution from countless possibilities, towards an objective function that we want to maximise or minimise [Orth et al. \(2010\)](#). The problem is a linear programming problem, which means all the reactions constraints are linear equations.

In FBA, we assume that the whole system is at a steady state so all the concentrations of metabolites stay constant. Whatever is produced by one reaction is immediately consumed by another one, so that nothing builds up. We present it mathematically as

$$SV = 0$$

where S is the stoichiometric matrix and V is the flux vector where each entry represents the rate of reaction in the system. Hence the production equals the consumption.

Because there are more reactions than metabolites in the model, there are many possible flux distributions that all satisfy the steady state constraint. We call this the feasible solution space. To improve the model, additional constraints are applied which are lower and upper bounds on reaction fluxes (v_{min} and v_{max})

$$v_{min,i} \leq v_i \leq v_{max,i}$$

which serve to either precise the reaction directionally or limit the maximum capacity of a specific reaction.

FBA uses a linear solver like simplex that can search through the feasible solutions space to find a flux vector(v) that maximizes or minimizes the objective function while satisfying all stoichiometric and

bound constraints. The optimisation problem is formulated like this

$$\max_v c^T v$$

where c is a vector of coefficients that defines the linear objective function (for example maximizing ATP production), S represents the stoichiometric matrix, and v the flux vector. (Figure 4)

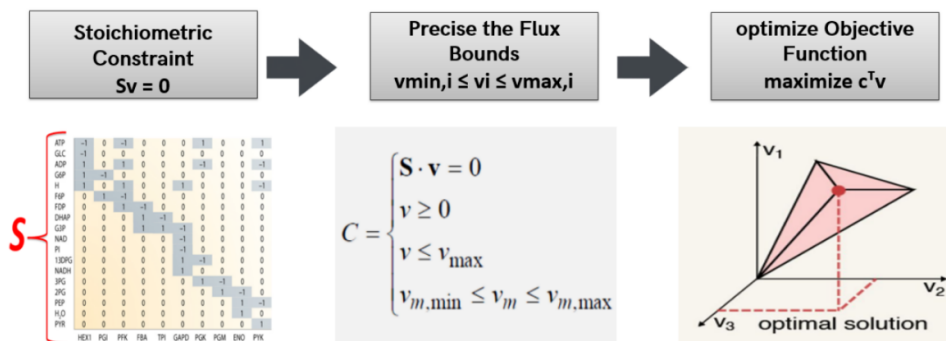


Figure 4: The flux balance analysis (FBA) linear programming framework.

1.2.4 E-Flux

FBA itself does not use gene expression data, it uses constraints from 0 to 1000 (or -1000 to 1000, in case of reversible reactions), so the model is not contextualized to a specific cell type or condition. We assume that reaction rates depend on the expression of the genes in the GPR rules, so the genes which code for the enzymes involved in the reaction. One method to constrain a metabolic network using expression data is E-Flux [Colijn et al. \(2009\)](#). E-flux integrates transcriptomics data to optimize reaction flux constraints. The maximal flux for each reaction is calculated as a function of the expression levels of the related gene

$$v_{max,i} = f(\{x_{g,i}\} \text{ for } g \in G_i)$$

where G is the set of genes encoding the enzymes for reactions i , and $x_{g,i}$ is the normalized expression level of a gene g .

Gene expression levels are first normalised in the range of 0.0 to 1.0 to be able to compare different conditions.

When we have multiple genes that encode for the same enzyme (AND relation) their expression is combined using the minimum value. On the other hand when they encode for iso-enzymes (paralogs) that can catalyze the same reaction (OR relation) we use the sum of them.(Figure 5)

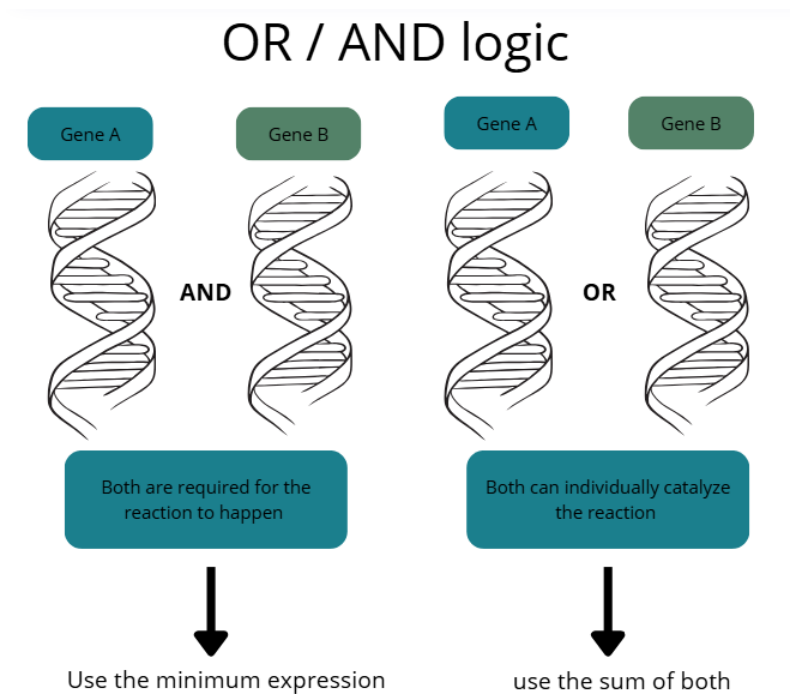


Figure 5: OR/AND gene logic applied in the E-Flux algorithm.

So combining FBA with E-Flux reflects the real enzymatic compositions of the cell, in different stages or states, like in my case different development states.

2 Aim of this Thesis

The aim of this thesis was to investigate how oxygen availability shapes mitochondrial metabolism in distinct cell types of the developing neocortex at E16.5 using genome-scale metabolic modeling. I integrated gene expression data with the mitoMammal metabolic model using the eFlux framework and applied cell type specific oxygen constraints reflecting the cortical oxygen gradient. Two objective functions cytosolic and mitochondrial ATP production were tested to identify how progenitors and neurons differ in their metabolic strategies. This approach reveals how the oxygen gradient drives the metabolic switch from glycolysis dependent progenitors to OXPHOS dependent neurons.

3 Materials and Methods

3.1 Data

The transcriptomic data used in this study originated from single-nucleus RNA sequencing of the developing murine neocortex at embryonic day E16.5. The dataset was generated and annotated by our collaborators from the Molecular Embryology group led by Prof. Tanja Vogel at the University of Freiburg, Germany.

For the metabolic analysis performed in my internship, I used pseudobulk expression profiles instead of the original single-cell data, the generation of the pseudobulk data was performed by Maelle spiller, as part of a parallel study done in the same laboratory. It was used because it reduces the technical noise compared to the single-cell sequencing data. The pseudobulk profiles were generated separately for each cell type, by aggregating expression information across cells while accounting for biological replicate variability. The resulting profiles represent the average expression pattern of each population, and served as the basis of metabolic models generated in my work.

3.2 MitoMammal as GEM

MitoMammal is a genome-scale metabolic model of mouse and human mitochondrial metabolism published by Chapman et al. [Chapman et al. \(2025\)](#). It was developed from MitoCore, a curated model of human cardiomyocyte mitochondrial metabolism [Smith et al. \(2017\)](#), by adding orthologue gene-protein-reactions rules for each reaction. The result is a model that simulates both mouse and human metabolism, by selecting the set of genes to use using the E-flux algorithm. This dual species design was the main contribution of the work, since no mouse specific mitochondrial model existed previously. The model contains 784 genes, 560 reactions and 446 metabolites across four compartments: the mitochondrial matrix, the cytosol, the intermembrane space, and the extracellular space. It covers all the

major pathways of mitochondrial metabolism, including the TCA cycle, the full ETC (CI–CV), fatty acid oxidation, the malate-aspartate shuttle, the glycine cleavage system, the proline cycle, and iron-sulfur cluster metabolism. One notable addition compared to MitoCore is the DHODH pathway, which links pyrimidine synthesis to CoQ reduction and had not been included in the original model.

3.3 Software and Computational Tools

All analyses were performed using Python 3.11 in Jupyter Notebook (VS Code). The mitoMammal metabolic model was analyzed using COBRApy for constraint-based reconstruction and analysis. Gene expression data was processed using pandas and numpy. The E-Flux integration framework was implemented to constrain reaction bounds based on enzyme abundance from pseudobulk transcriptomic data.

3.4 FBA Methodology

I integrated pseudobulk expression profiles into the mitoMammal model using the E-Flux algorithm to constrain reaction bounds based on gene expression levels. For each of the four cell types (apical progenitors, basal progenitors, deep layer neurons, and upper layer neurons), I performed flux balance analysis (FBA) on the constrained models for each cell type with biologically appropriate oxygen constraints reflecting their position in the developing cortex. I constrained the oxygen transport (O2t reaction) to different maximum values: apical progenitors (3), basal progenitors (6), deep-layer neurons (9), and upper-layer neurons (12). I swept the oxygen and observed its saturation levels for each cell type.

I performed FBA using two different objective functions: the first one is cytosolic ATP production (OF_ATP_mitoMap) and the second one mitochondrial ATP synthesis (ATPtmB_mitoMap). I applied parsimonious FBA (pFBA) to minimise total metabolic flux, and select the most biologically meaningful solution among all the optimal solutions.

4 Results and Discussion

4.1 Oxygen constraint selection

To investigate how oxygen availability constrains OXPHOS in each cell type, and to get oxygen constraints for my models. I performed an oxygen analysis by varying oxygen uptake levels from 1 to 19 and tracking complex V (ATP synthases) flux. All the cells showed a linear increase in ATP synthesis, which is expected because OXPHOS is oxygen dependent. However, I observed different oxygen capacities among cell types: progenitors (AP and BP) reached a plateau at lower oxygen levels (107 and 93) compared to neurons (DL and UL) which continued to increase towards higher plateaus (144 and 137).

This behaviour suggests that progenitors have a functional ETC but they are probably constrained by other metabolite constraints. This is consistent with in vitro observations showing that progenitors maintain mitochondrial efficiency under hypoxia conditions [Lages et al. \(2015\)](#).

Based on where each cell type reaches the plateau, I chose the oxygen uptake constraints also reflecting their position in the developing cortex and putting in consideration the oxygen gradient that exists there. Knowing that the fetal brain has low oxygen at 1-6% PO₂, much lower than the adult brain which is at 20%, I used 18 as the maximum oxygen uptake, because that's where all cell types reached their maximum OXPHOS capacity. Then I assigned different constraints to each cell type based on their location: apical progenitors reside in the most hypoxic part (VZ) so I restricted oxygen to 3 mmol/gDW/hr. Basal progenitors are in the SVZ, which is less hypoxic, so I set oxygen levels to 6 mmol/gDW/hr. DLs are closer to the surface where blood vessels start growing, so here, I restricted oxygen to 9 mmol/gDW/hr. ULs are in the most oxygenated regions, so here I chose the highest constraint of oxygen at 12 mmol/gDW/hr.

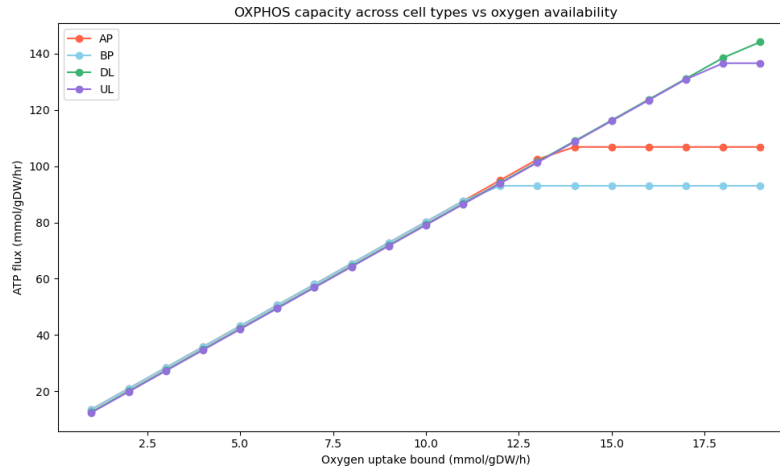


Figure 6: Oxygen constraint selection. OXPHOS capacity (Complex V flux) across cell types versus oxygen availability.

4.2 ATP Production Levels: Cytosolic vs. Mitochondrial

After optimising the model with the chosen oxygen constraints, for cytosolic and mitochondrial ATP production, I found different ATP production levels across all cell types: progenitors (AP and BP) made very little ATP amount in the cytosol (26.60 and 34.34) but they produced a lot of it in the mitochondria (132.44 and 150.53). Neurons (DL and UL) on the other hand produced relatively large amounts of ATP on the cytosol (54.62 and 40.37). DLs specifically had the lowest amount of mitochondrial ATP (115), and ULs had the highest (152.27). This difference in ATP compartmentalization reflects the metabolic constraints imposed by oxygen availability. Progenitors, constrained to low oxygen levels (3-6 mmol/gDW/hr), rely on less efficient ATP production pathways and accumulate ATP primarily in the mitochondrial compartment. In contrast, deep-layer and upper-layer neurons, with access to higher oxygen levels (9-12 mmol/gDW/hr), can utilize more efficient oxidative phosphorylation pathways, resulting in higher cytosolic ATP production. The higher OXPHOS efficiency in oxygenated environments drives the metabolic differences between these cell types, regardless of where ATP is synthesized or consumed [Komabayashi-Suzuki et al. \(2019\)](#).

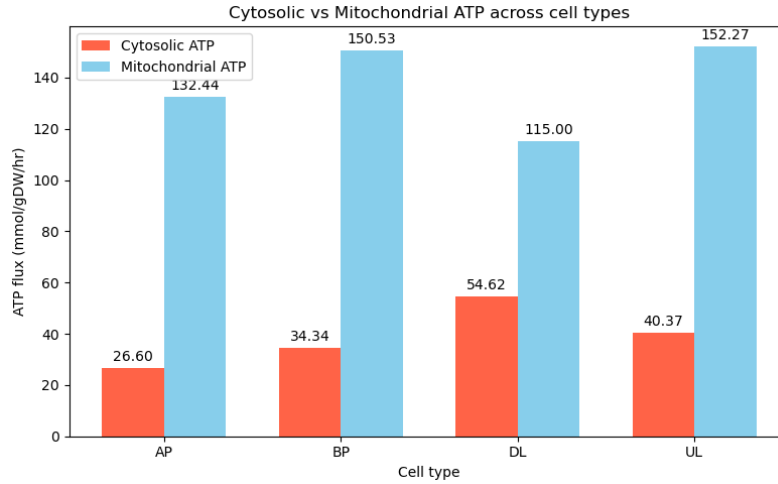


Figure 7: Cytosolic versus mitochondrial ATP production across cell types.

4.3 ATP Source Decomposition: Glycolysis, TCA Cycle, and OXPHOS

In order to investigate further, which reactions produce ATP in both cytosol and mitochondria, I analysed how ATP was produced between three different pathways: glycolysis (PYK + PGK), the TCA cycle (SUCOASm), and OXPHOS (CV_mitoMap), always using the same oxygen constraints I assigned for each cell type.

Progenitors exhibited a very different ATP production profile compared to neurons. AP obtained most of its ATP from the TCA cycle (56.6%), with an important contribution from glycolysis (28.3%) but

minimal OXPHOS (15.1%). BP as well showed a very similar profile, with TCA cycle providing (47%) ATP, glycolysis (29%) and a bit more OXPHOS than AP (24.1%).

Neurons on the other hand demonstrated a huge metabolic shift towards OXPHOS. DLs increased their OXPHOS contribution to (40.6%) while completely reducing the TCA cycle to (13.6%), but glycolysis remained still very active with the highest percentage among all cell types (45.9%). ULs had the highest OXPHOS activity at 45.3% with moderate TCA cycle activity (22.4%). Notably, TCA cycle activity remained elevated in progenitors even under the restricted oxygen conditions, suggesting that glutaminolysis and other anaplerotic pathways sustain the TCA cycle independently of glycolysis-derived pyruvate [Namba et al. \(2020\)](#). This ATP production profile indicates a clear developmental shift: progenitor cells rely predominantly on TCA cycle and glycolysis for ATP generation, whereas differentiated neurons increasingly use OXPHOS as their primary energy source [van den Ameele and Brand \(2019\)](#). This metabolic rewiring likely reflects both the increased oxygen availability in more superficial cortical layers and the elevated energetic demands of differentiated neuronal cells compared to proliferating progenitors.

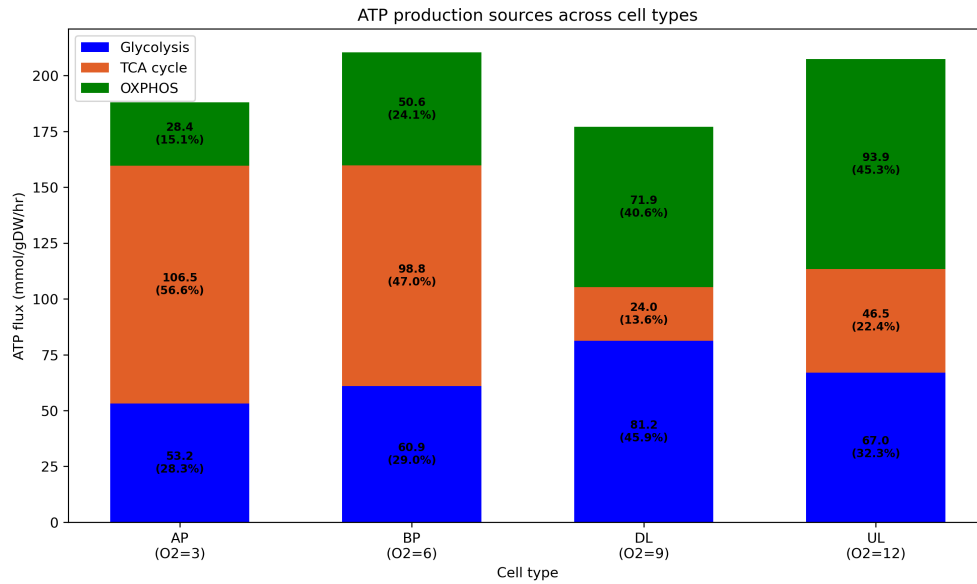


Figure 8: ATP production sources across cell types. Stacked bar chart showing percentage contributions from glycolysis (blue, PYK+PGK), TCA cycle (orange, SUCOASm), and OXPHOS (green, CV_mitoMap).

4.4 ETC Complex Flux Analysis

To investigate even further, I looked at how much flux goes through each ETC complex when the model is optimizing cytosolic ATP. The results show that progenitors and neurons use different amounts of ETC. Progenitors had lower flux through Complex I (6.00 and 12.00) compared to neurons (18.00 and 24.00).

Complex II showed no activity in any cell type. Complex III and IV also increased from progenitors to neurons, reflecting the overall pattern. Complex V, which is ATP synthase and actually makes the ATP, showed the biggest difference: progenitors had a flux of (28.36 and 50.59) while neurons had a flux of (71.89 and 93.88). The progression of flux through the ETC, with Complex I as the main entry point, increases progressively across developmental stages, indicating that neurons have higher capacity for OXPHOS driven ATP synthesis compared to progenitors. Although Complex I is known to be essential for progenitor proliferation and differentiation [Cabello-Rivera et al. \(2019\)](#); [van den Ameele and Brand \(2019\)](#), the low flux through Complex I in progenitors reflects their oxygen limited microenvironment rather than the absence of a functional electron transport chain in the mitochondria.

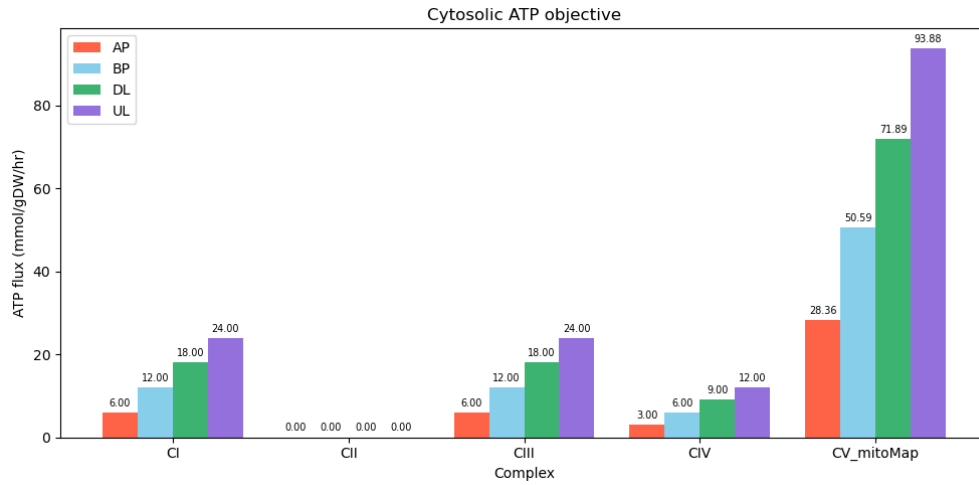


Figure 9: Electron transport chain complex flux analysis. Bar chart showing absolute flux (mmol/gDW/hr) through ETC complexes I-V when optimizing cytosolic ATP production.

5 Conclusion

This study investigated metabolic pathway changes during neural development, by modeling mitochondrial metabolism in four cell types of the developing neocortex using flux balance analysis, with pseudobulk gene expression data, and physiologically constrained oxygen levels. I found a clear shift in ATP production levels, with increasing usage of OXPHOS in developed neurons.

Progenitor cells (apical and basal progenitors) rely dominantly on the TCA cycle and glycolysis for ATP generation, with a very small contribution from OXPHOS. This metabolic strategy is due to their hypoxic microenvironment (ventricular and subventricular zones), where oxygen availability is very limited.

On the other hand neurons have a different metabolic strategy. Deep layer and upper layer neurons shift towards OXPHOS, where 40-50% of their ATP comes from the electron transport chain. This change happens because they live in more oxygenated layers. Using different energy production pathways gives

them probably also the possibility to respond to other cellular needs, such as migration, growing axons, and making synapses.

The key finding is that oxygen availability is the main factor that controls when progenitors switch to OXPHOS. Progenitors have the ETC functional but can't use it efficiently in their hypoxic environment, so they rely on less efficient, oxygen independent pathways instead. This metabolic flexibility allows progenitors to survive and proliferate in low oxygen zones. When neurons form in more oxygenated areas, they can finally use the most efficient pathway for ATP production.

This work demonstrates that constraint based metabolic modeling reveals how different cell types use different ATP production pathways, and highlights why the oxygen gradient in the developing brain is important for neural development.

5.1 Limitations of This Study

This work has several limitations. First, the true metabolic objective of these cells is unknown. I assumed ATP production, but cells may optimize for multiple goals. Second, I only considered mitochondrial metabolism and ignored other important pathways like the pentose phosphate pathway and one-carbon metabolism. Third, these are computational predictions that need experimental validation through direct metabolic measurements such as oxygen consumption rates, lactate production rates, glucose utilization rates, and mitochondrial membrane potential. Finally, the analysis is limited to a single timepoint (E16.5), so I don't know how metabolic states change during development.

6 Future Perspectives

Moving forward, I will be using an algorithm (developed by Jean-Marc and Kate Meeson at Manchester University) to further improve our FBA results. This algorithm uses a simulated annealing approach to identify the most likely objective function of a cell, based on gene expression data and experimentally determined reactions that are active or inactive. Rather than finding one optimal solution, it returns the probability of each of the user-proposed objective functions and thus allows us to estimate what the most likely objective function of the cell will be.

I will run this algorithm on the same cell types and compare its predictions with my current FBA results. This will help decide whether the metabolic strategies I identified with FBA match with what the cells' most likely objective functions.

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